



HACKATHON PRESENTATION

PEDAGOGY

Centralized Technical Event Management System

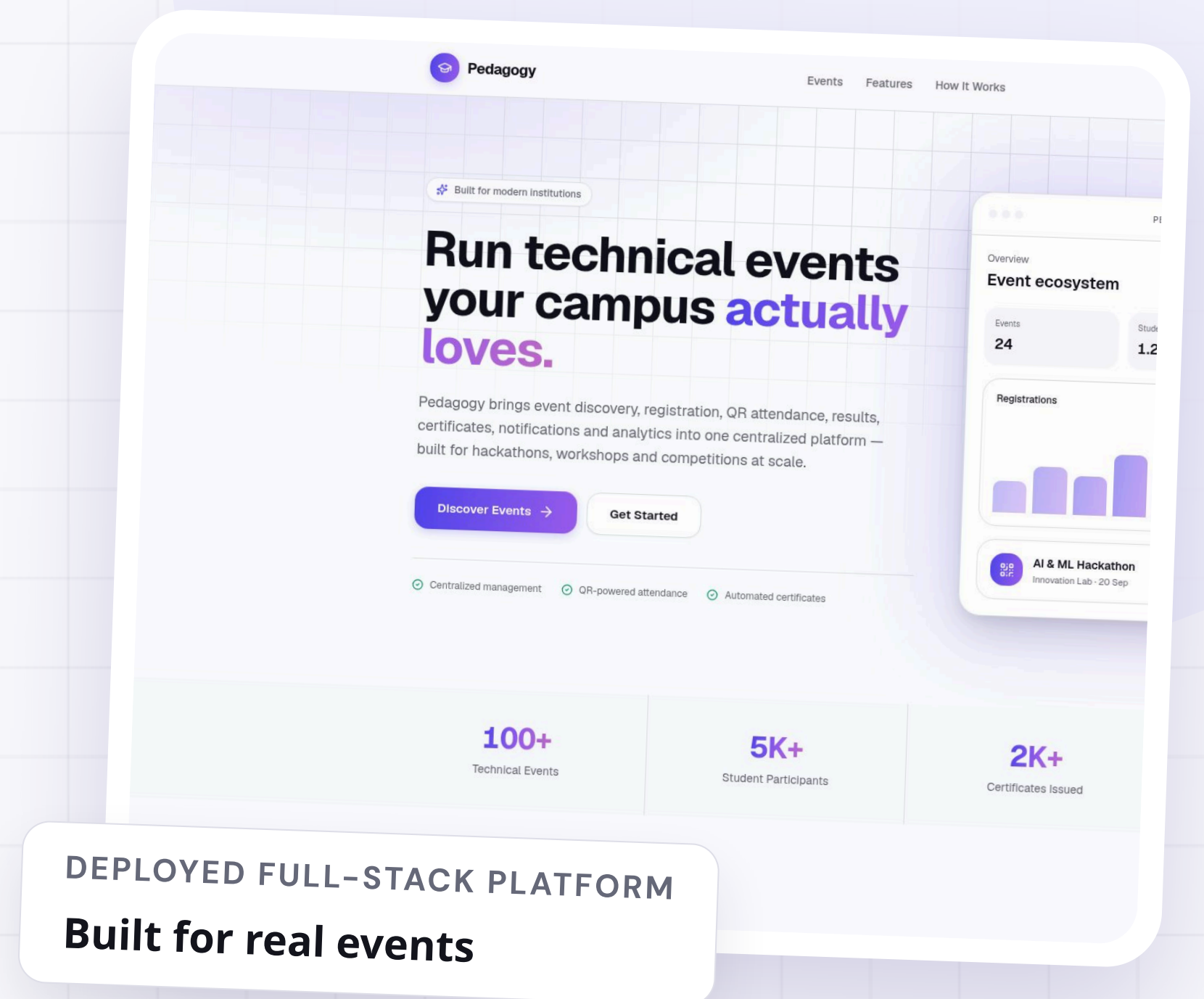
One platform. Every event. Every stakeholder.

LIVE DEMO pedagogy-plum.vercel.app

GITHUB github.com/VarunMehta278/pedagogy

TEAM ID 23

TEAM MEMBERS Dewik Bavishi, Varun Mehta

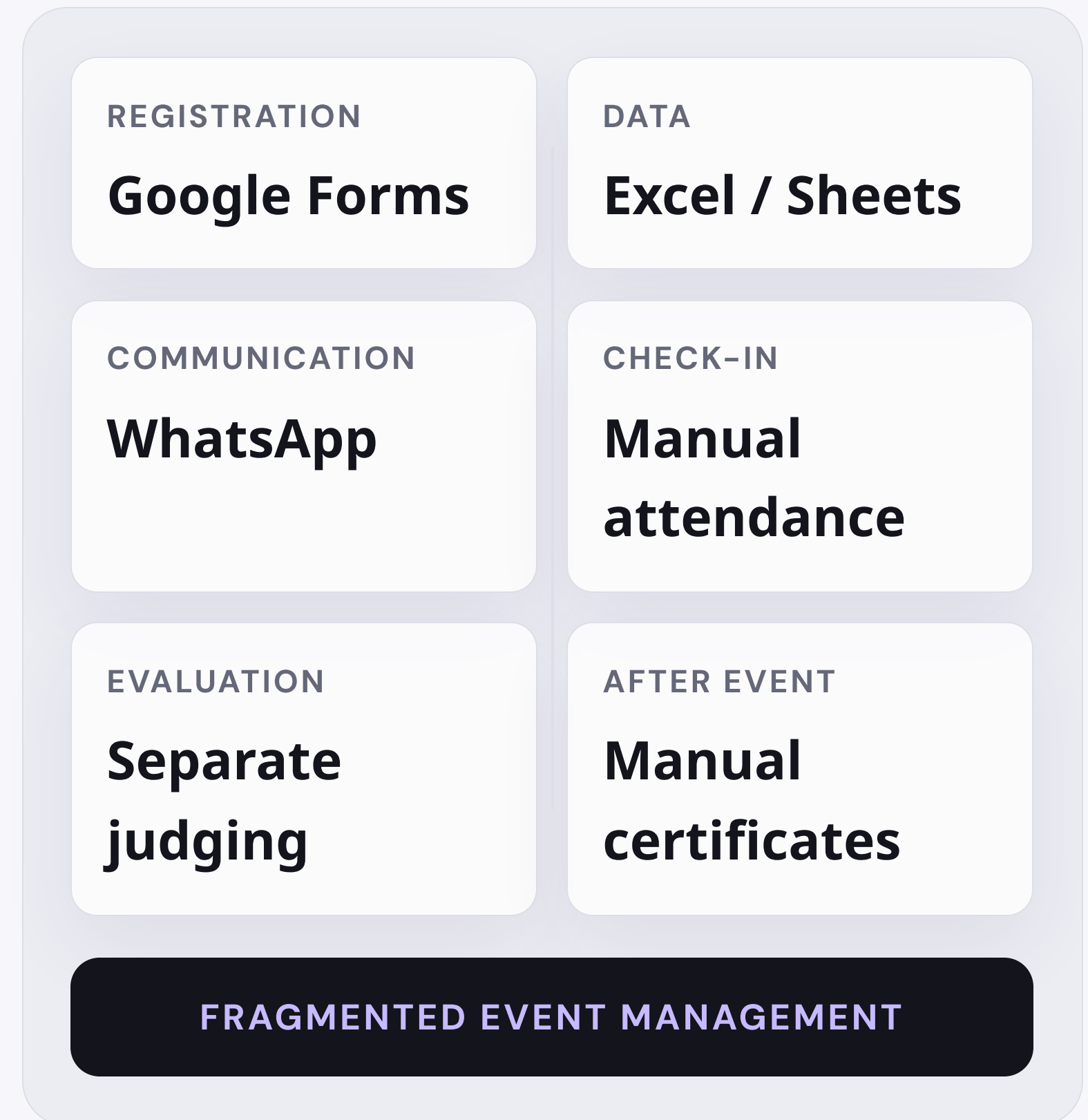


01 / FRAGMENTATION

Technical events are managed across too many tools

One event can require several disconnected tools just to manage the basics.

- Registrations become difficult to track
- Participant data gets scattered
- Attendance is often manual
- Judges use disconnected evaluation processes
- Results and certificates need additional coordination



02 / CENTRALIZE

Meet Pedagogy

A single platform for the complete technical-event lifecycle.

Instead of connecting several tools around an event, Pedagogy connects the entire workflow inside one system.



ONE PLATFORM REPLACING FRAGMENTED WORKFLOWS

EVENTS → REGISTRATION → TEAMS → ATTENDANCE → JUDGING

03 / SCOPE

Everything an event needs. In one platform.

01 / EVENT MANAGEMENT

Create · publish · manage

Registration deadlines
Participant limits

02 / PARTICIPATION

Register · form · join

Team creation and joining
Registration QR codes

03 / OPERATIONS

Check in · verify · assist

QR attendance
Volunteer workflows

04 / JUDGING

Assign · evaluate · score

Remarks
Multi-judge workflows

05 / OUTCOMES

Results · certificates

Winner and participation
Digital verification

06 / MANAGEMENT

Notify · manage · analyze

User management
Event analytics

04 / ROLE-BASED ACCESS

Five roles. One connected system.

ST 01

Student

- Discover events
- Register and form teams
- Access QR
- View results
- Download certificates

FO 02

Faculty / Organizer

- Create events
- Manage participants
- Track attendance
- Manage results
- Send announcements

J 03

Judge

- View assigned events
- Evaluate participants
- Enter scores
- Add remarks
- Submit evaluations

V 04

Volunteer

- View assigned events
- Verify participants
- Scan QR codes
- Assist with attendance

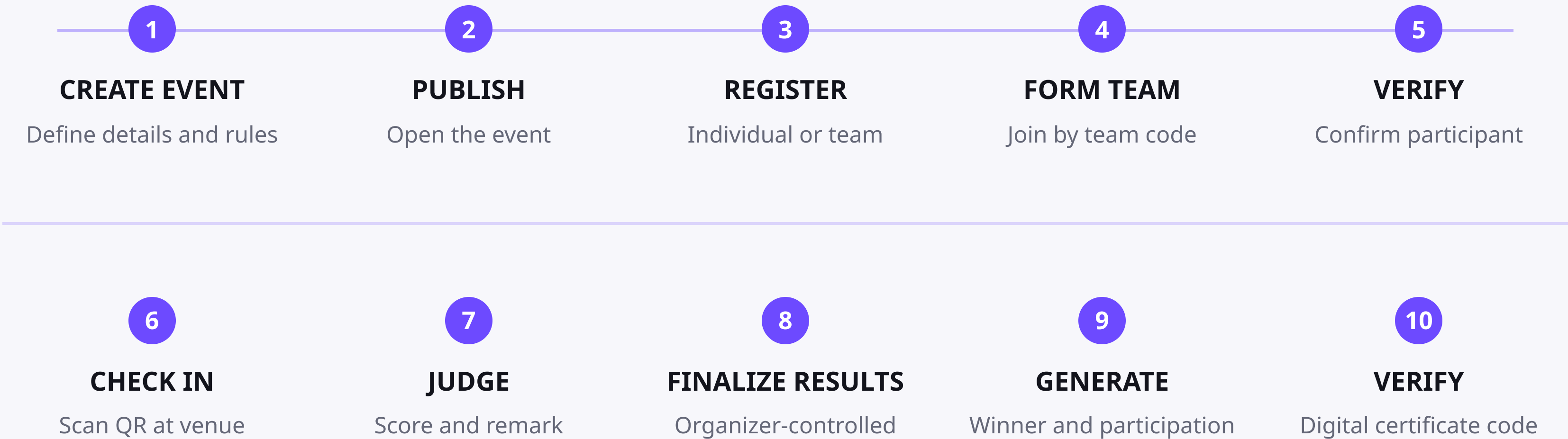
AD 05

Admin

- Manage users
- Manage events
- View analytics
- Monitor the platform

05 / END TO END

From registration to certificate



Registration → team → attendance → judging → results → certificates

06 / BUILT FOR REAL EVENTS

Teams + QR attendance

TEAM MANAGEMENT

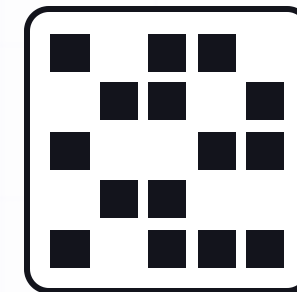
01

Create team → Share code → Members join

- Individual or team-based events
- Team leader and member management
- Minimum / maximum team size
- Team-based judging, results and certificates

QR-BASED ATTENDANCE

02



Registration QR
→ scan at venue
→ attendance recorded

- Fast participant verification
- Duplicate attendance protection
- Organizer / volunteer scanning
- Attendance linked to registration

07 / BEYOND ATTENDANCE

The event doesn't end at attendance.

01

Judging

- Assigned judges
- Participant / team evaluation
- Scores and remarks
- Multiple independent evaluations

02

Results

- Position and score
- Remarks
- Organizer-controlled results
- Individual or team results

03

Certificates

- Winner certificates
- Participation certificates
- PDF generation
- QR-based verification

JUDGING → RESULTS → CERTIFICATES

08 / UNDER THE HOOD

How Pedagogy works under the hood

FRONTEND / VERCEL

Next.js · TypeScript · Tailwind CSS · shadcn/ui



BACKEND / RENDER

Node.js · Express.js · TypeScript · REST APIs



DATABASE / SUPABASE

PostgreSQL

SERVICES

JWT authentication

**bcrypt password
hashing**

**QR generation /
scanning**

PDF certificates

Notifications

Analytics

USERS

Student

Faculty

Judge

Volunteer

Admin

Users → Next.js frontend → Express REST API → Auth + authorization → Supabase PostgreSQL

09 / WHAT WAS ACTUALLY HARD?

The hard parts were in the handoffs.

01 / Multi-role authorization

Different roles require different permissions.

Server-side auth + role-based authorization

02 / Team participation

Individual and team events stay consistent across the workflow.

03 / Event-day operations

QR attendance must be fast while preventing duplicate check-ins.

04 / Multi-judge evaluation

Independent evaluations, organizer-controlled final results.

05 / Production authentication

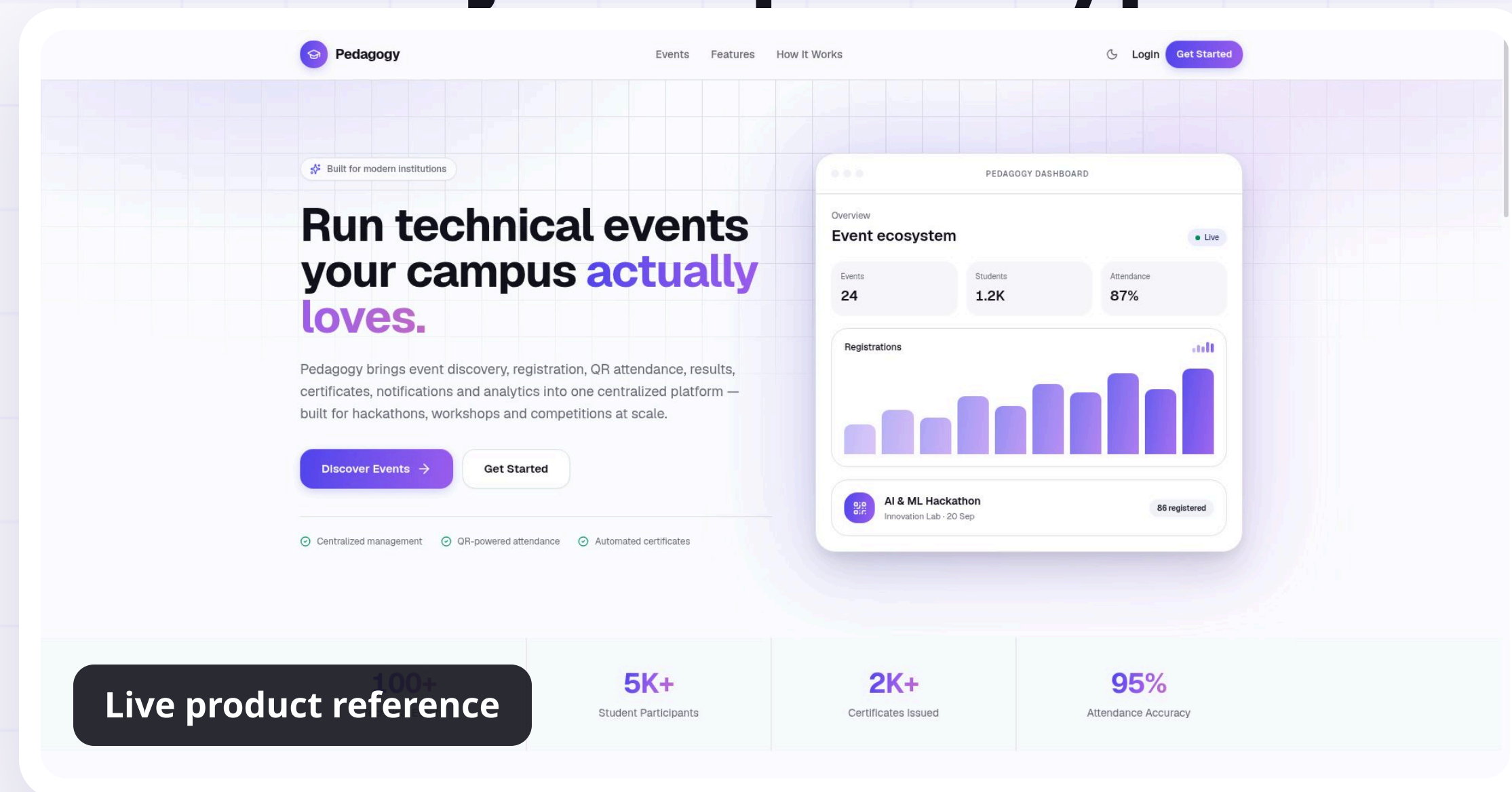
The Vercel frontend and Render backend required HTTP-only secure cookies and careful CORS configuration.

SECURITY LAYER

- JWT authentication
- HTTP-only secure cookies
- bcrypt password hashing
- Protected APIs and CORS
- Login rate limiting
- Database constraints + validation

10 / THE LIVE PRODUCT

This isn't just a prototype.



DEPLOYED

**Built, tested, and
deployed as a working
full-stack platform.**

Frontend

Vercel

Backend

Render

Database

Supabase

LIVE DEMO

pedagogy-plum.vercel.app

GITHUB

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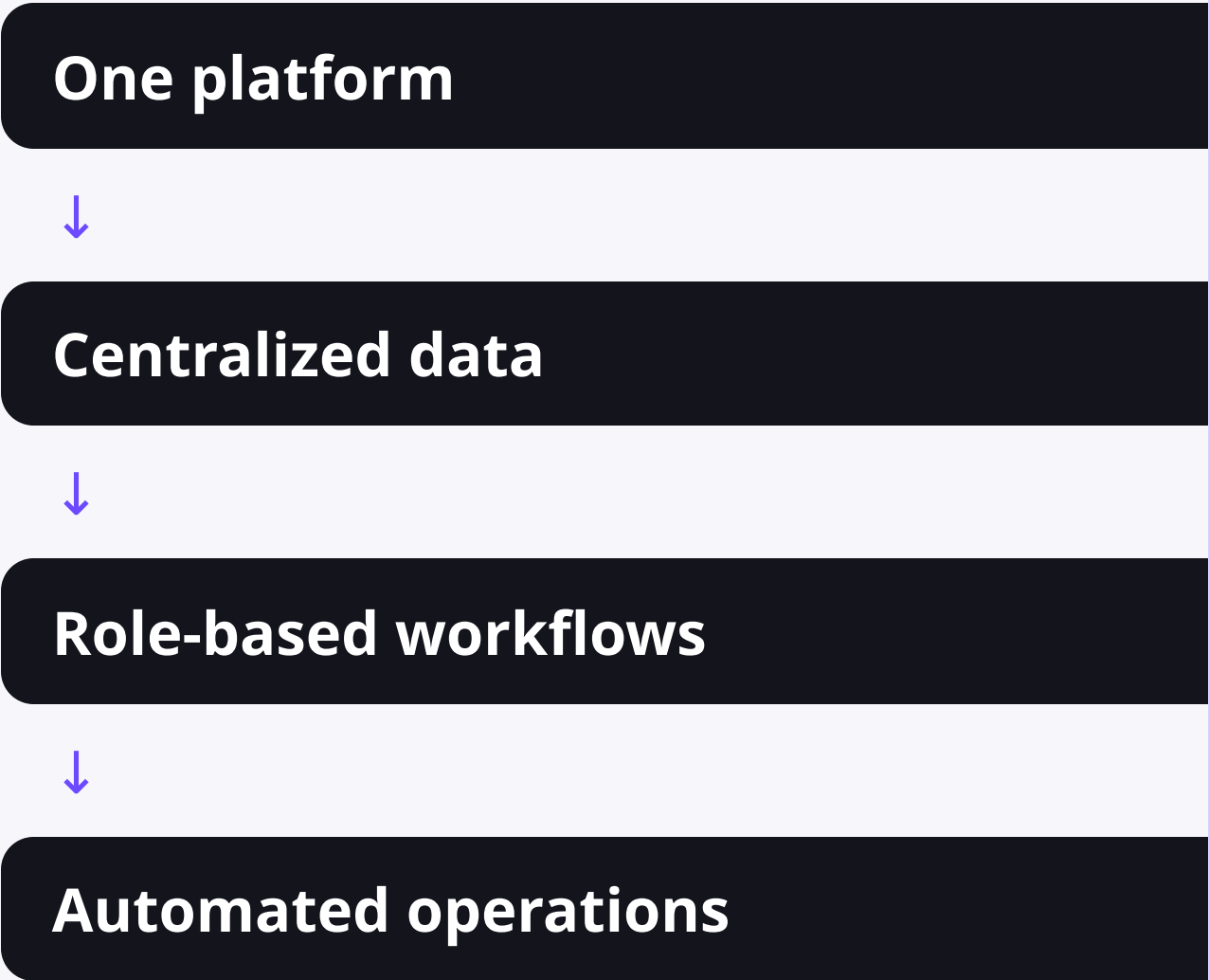
11 / WHAT THIS ENABLES

From fragmented workflows to one event ecosystem

BEFORE PEDAGOGY



WITH PEDAGOGY



FUTURE SCOPE

- Multi-institution SaaS
- Mobile application
- Push notifications
- Payment-enabled events
- Advanced reporting
- ERP + calendar integrations

Pedagogy can evolve into a complete institutional event ecosystem.

THANK YOU

One platform. Every event.

Every stakeholder.

CONTINUE THE DEMO

LIVE DEMO

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TEAM Underdogs

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